Downsampling

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Key Concepts

1) Downsampling reduces the sampling rate by an integer factor M.

    a) If the original sampling interval is T, the down sampled sampling interval is MT.

    b) If the original sampling frequency Ω_s, the down sampled sampling frequency is Ω_s/M.

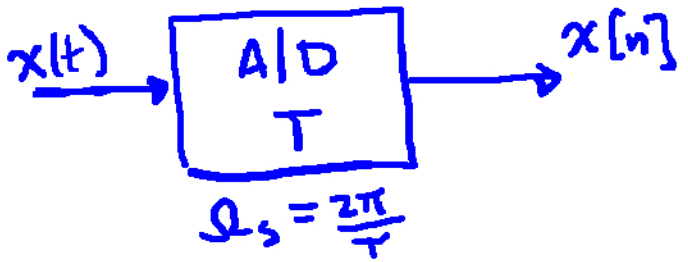
2) Discarding all samples except every M^{th} sample reduces the sampling rate, but introduces aliasing if the original signal is not band limited to $|\omega| < \frac{\pi}{M}$ rads.

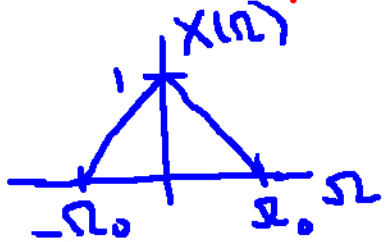
3) Downsampling is performed in practice by first applying a discrete-time lowpass filter with stop band $|\omega| > \frac{\pi}{M}$ followed by the step of discarding all samples except every M^{th} one.

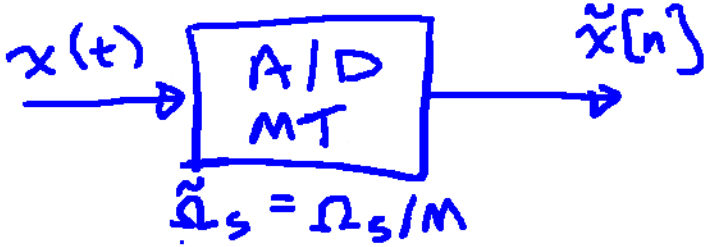
4) Downsampling stretches the frequency content of the original signal in $|\omega| < \frac{\pi}{M}$ to occupy the entire band $-\pi < \omega \leq \pi$ and reduces the amplitude by a factor of M.

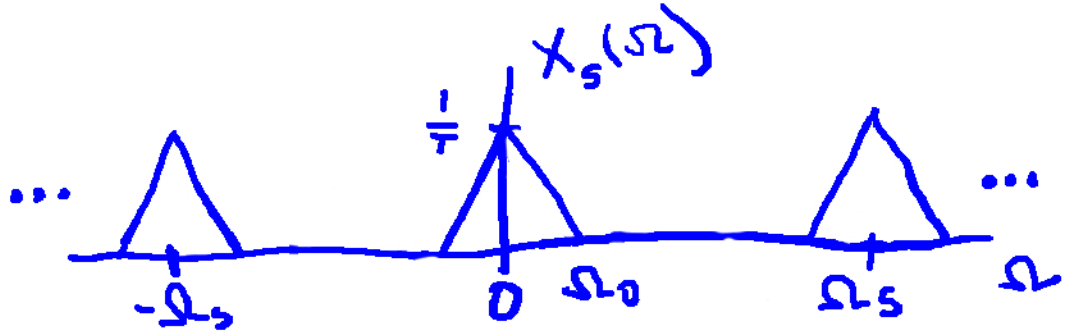
Video Screenshots

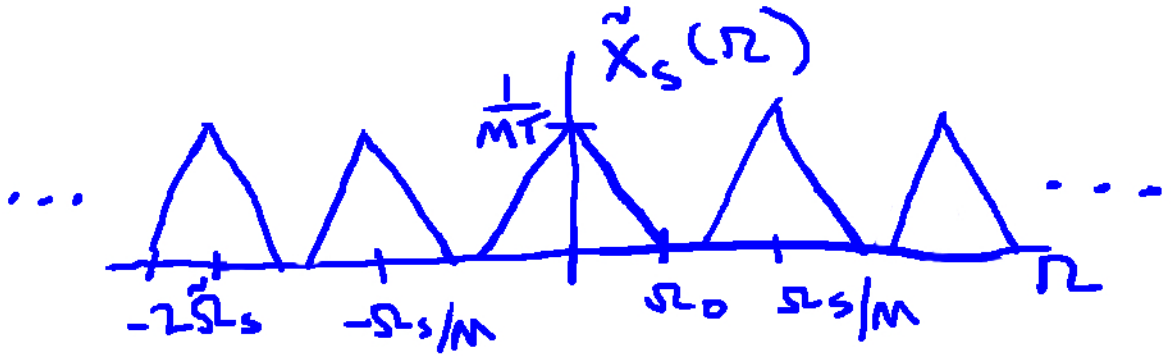
Downsampling



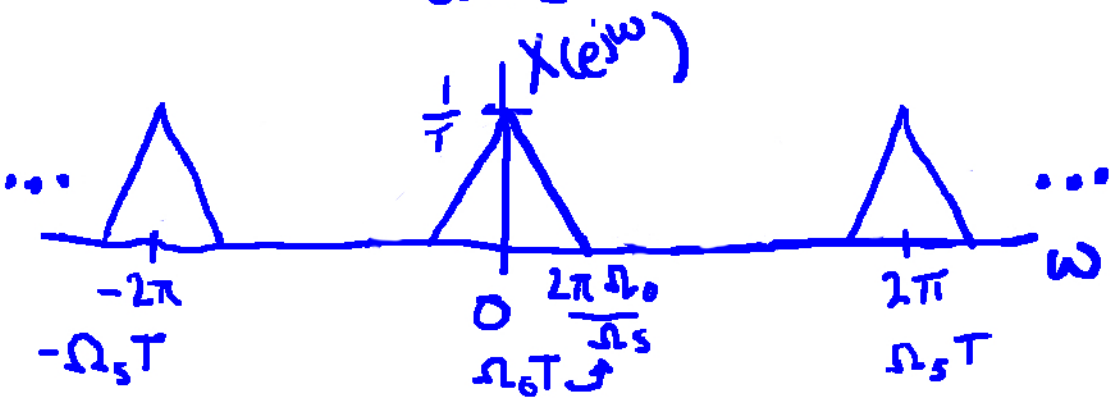


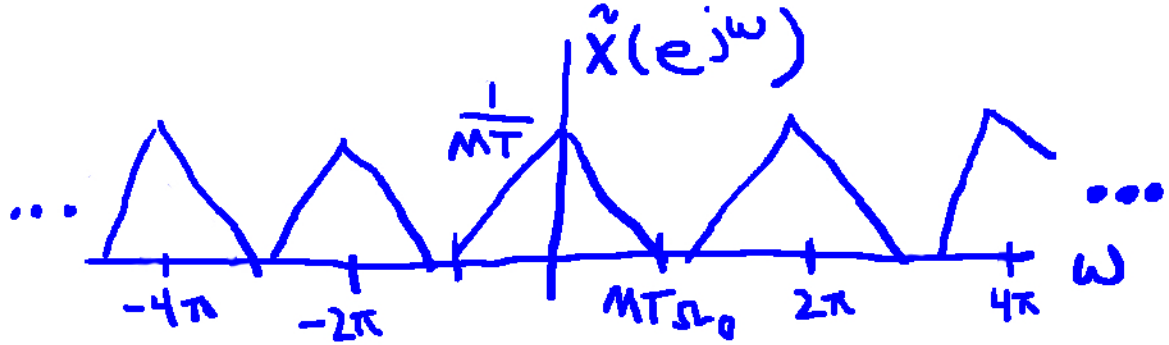






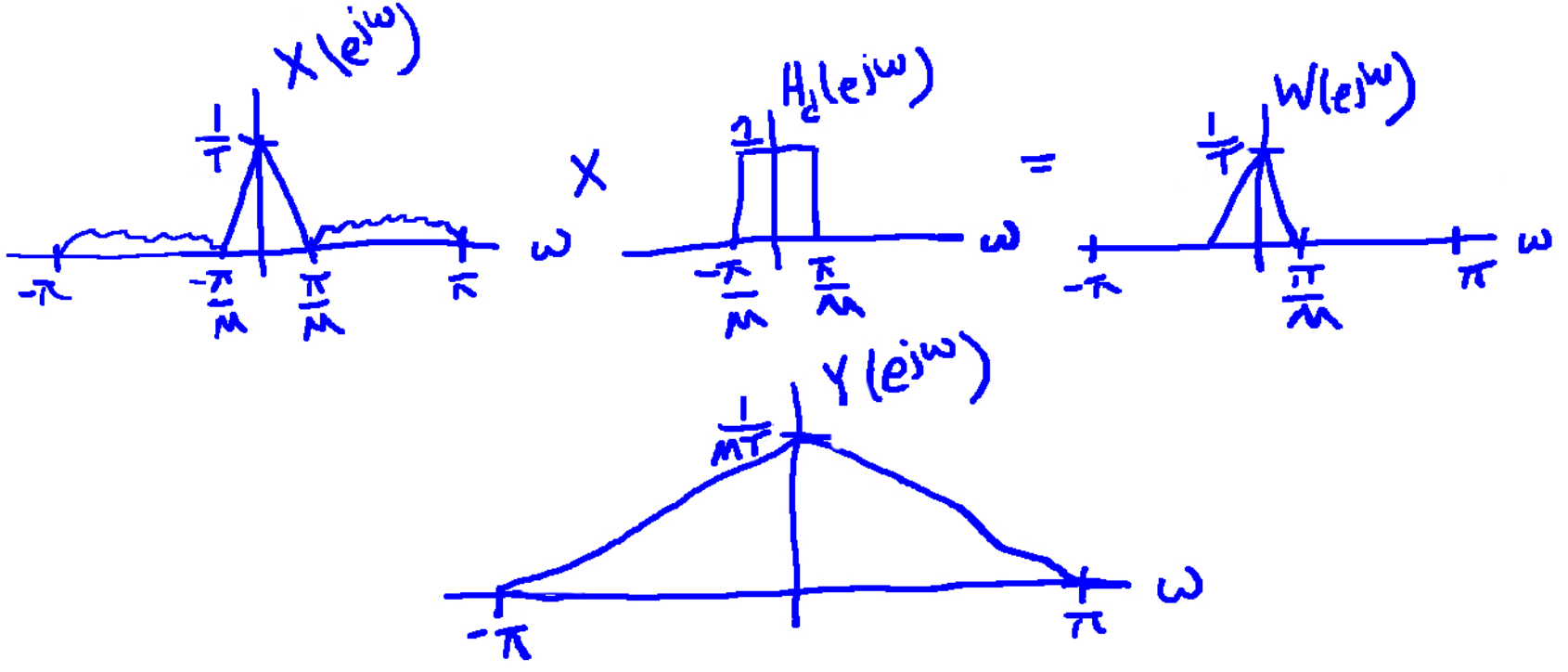
$\omega = \Omega T$





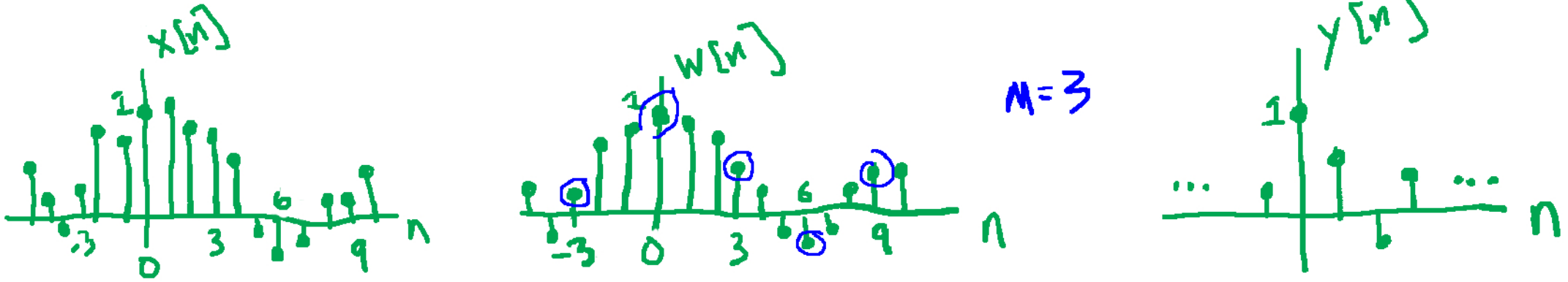
$\frac{\Omega_s}{M} \cdot MT = 2\pi$





Note: if $y[n] = w[Mn]$, then $Y(e^{j\omega}) = \frac{1}{M} \sum_{m=0}^{M-1} W(e^{j(\omega - m2\pi)/M})$


x[n]
Low Pass
$H_d(e^{j\omega})$
w[n]
Discard values
$n \neq Mk$
y[n] = w[Mn]


x[n]
w[n]
M=3
y[n]
n


x[n]
Low Pass
π/M
w[n]
↓ M
y[n] = w[Mn]

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